Lab 2 : Implementing FTP Protocol

What is FTP and What is it used for ?

FTP (File Transfer Protocol): is a network protocol for transmitting files between computers over Transmission Control Protocol/Internet Protocol (TCP/IP) connections. Within the TCP/IP suite, FTP is considered an application layer protocol.
In an FTP transaction, the **end user's computer** is typically called the **local host.** The second computer involved in FTP is a **remote host,** which is usually a **server**. Both computers need to be connected via a network and configured properly to transfer files via FTP. Servers must be set up to run FTP services, and the client must have FTP software installed to access these services.

FTP is a client-server protocol that relies on two communications channels between the client and server: a command channel for controlling the conversation and a data channel for transmitting file content.

Here is how a typical FTP transfer works:

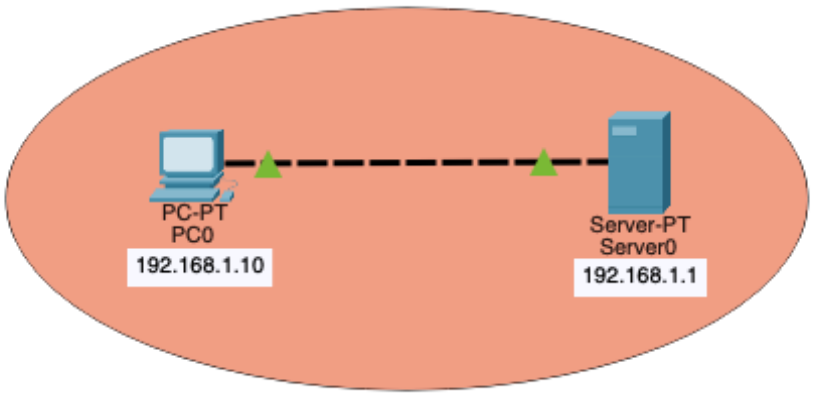
1. A user typically needs to log on to the FTP server, although some servers make some or all of their content available without a login, a model known as anonymous FTP.
2. The client initiates a conversation with the server when the user requests to download a file.
3. Using FTP, a client can upload, download, delete, rename, move and copy files on a server.

FTP sessions work in active or passive modes:

- **Active mode:** After a client initiates a session via a command channel request, the server creates a data connection back to the client and begins transferring data.
- **Passive mode:** The server uses the command channel to send the client the information it needs to open a data channel. Because passive mode has the client initiating all connections, it works well across firewalls and network address translation gateways

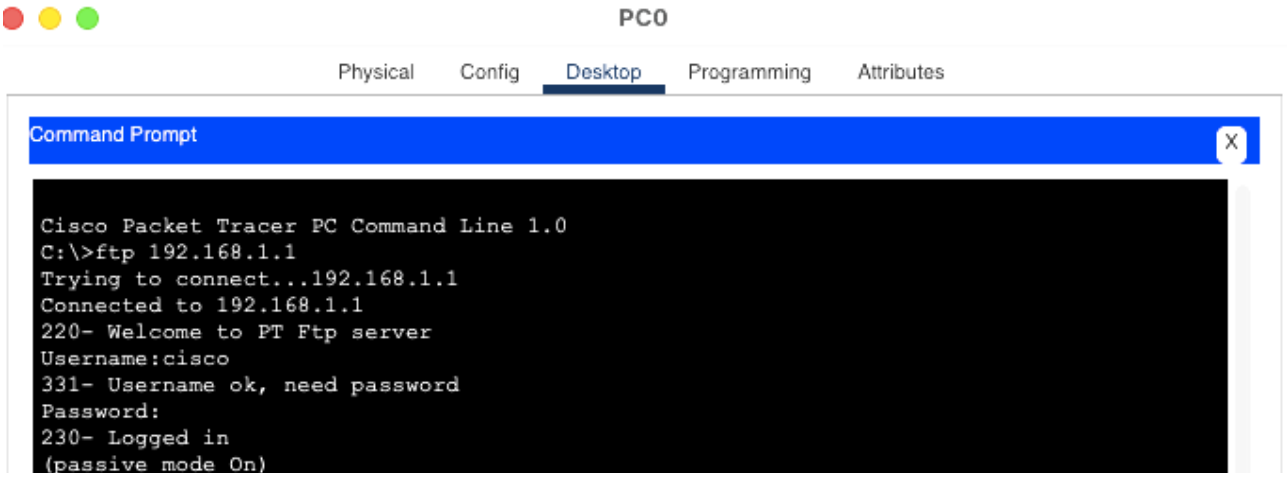
Topology :

Step 1: Draw this topology & Assign these IPs for each Client and Server.



Step 2: Now try using an FTP client built in the PC to send files to an FTP server configured in the Server.
From the PC's command prompt, FTP the server using the server IP address by typing:
ftp 192.168.1.1

Step 3: Enter the username & Password **cisco** (Default), you are now in FTP prompt.



Step 4: Create a file in the Laptop then upload it to the server using FTP.
To do this, open the Text Editor in the PC, create a file and give it your name of choice.
Type any text in the editor then save your file. For Example: FTP_test.txt

Step 5: Now upload this file on the Server by typing:
put FTP_test.txt



Step 6: Once file upload is successful, go to the Server FTP directory to verify if the file sent has been received. To do this, go to Server—> Services—>FTP. Here look for FTP_test.txt sent from the PC.

To check **other FTP commands** supported by the FTP client running on the Laptop(or PC), you can use a question mark (?) on the PC's command prompt as shown below:

